# Jin Tao

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## Education

**Technische Universität Berlin** — Berlin, Germany
*M.Sc. in Computational Engineering Science; GPA: 1.3 (1.0 best)* — *Feb. 2025 – Present*

**Technische Universität Berlin** — Berlin, Germany
*M.Sc. in Vehicle Engineering* — *Sep. 2020 – Feb. 2025*

**Hainan University** — Hainan, China
*B.Sc. in Vehicle Engineering* — *Sep. 2015 – June 2019*

## Experience

**Researcher (previously Working Student)** — May 2026 – Present
*Center for Tangible Artificial Intelligence and Digitalization (ZEKI)* — *Berlin, Germany*

- Develop a full-stack robotics platform on TurtleBot integrating SLAM, navigation, semantic mapping, autonomous exploration, and vision-language models, with an AI agent that interprets user requests into robot actions.
- Develop a multi-robot collaboration system for disaster response, integrating UGVs and UAVs for collaborative mapping and task coordination through a central device.

**Research Assistant** — Oct. 2025 – Apr. 2026
*Robotics and Biology Laboratory (RBO)* — *Berlin, Germany*

- Designed and built an acoustic gripper from scratch, using acoustic signatures to identify grasped objects.
- Improved the software pipeline for a Franka manipulator by resolving a persistent impedance-controller bug that caused abnormal oscillations at higher impedance settings.

**Working Student** — Sep. 2024 – Oct. 2025
*Mbition GmbH* — *Berlin, Germany*

- Developed a full-stack software system for monitoring test vehicles and managing vehicle bookings.

**Software Engineer** — Oct. 2022 – Mar. 2024
*TU Berlin Formula Student Team - FaSTTUBe* — *Berlin, Germany*

- Developed a ROS 2 trajectory-tracking and control package using local and global planned trajectories, with pure pursuit and model predictive control (MPC), targeting minimum lap time.
- Developed a steering control system using cascaded current, speed, and position PID loops to achieve fast, stable tracking of commanded steering positions.

**Team Leader** — May 2018 – Nov. 2019
*Hainan University Formula Student Team* — *Hainan, China*

- Designed and built the vehicle's double-wishbone suspension, applying vehicle dynamics principles to determine its geometry.
- Led team organization and coordination, and competed as the endurance-event driver at Formula Student China 2019.

## Achievements and Projects

**Magnetic Joint-State Estimation** — Mar. 2026 – Sep. 2026

- Proposed and developed a ball-joint orientation estimation system using two Adafruit magnetic sensors costing less than €7 each and a magnet to estimate yaw, pitch, and roll.
- Evaluated orientation accuracy against an Xsens MTi-630 IMU, achieving comparable accuracy under the tested conditions without long-term drift.
- Preparing a conference paper with the project team.

**Machine Learning for Visuo-Tactile Sensor Fusion** — Mar. 2026 – Sep. 2026

- Developed training and evaluation infrastructure for estimating in-hand object pose from RGB images and tactile measurements, using ground-truth pose annotations.
- Implemented and compared ten existing methods, then developed a transformer-based fusion model that matched the best evaluated baseline's pose-estimation accuracy.

**Master's Thesis: Autonomous Shuttle Pedestrian Following** Aug. 2024 – Jan. 2025

- Developed and deployed a full-stack pedestrian-following system on a real autonomous shuttle, integrating YOLO-based gesture detection with commands to stop, resume driving, and adjust following distance.

**GPU-Accelerated Visual SLAM and Navigation Integration** Mar. 2025 – Sep. 2025

- Deployed GPU-accelerated visual SLAM on a Jetson Xavier-based mobile robot and integrated its outputs with Nav2 by projecting 3D map points into a 2D occupancy grid.
- Reduced CPU utilization by 40% while increasing GPU utilization by 20%, and improved navigation through faster map–odometry updates and more accurate localization.

## Skills

**Languages**: English (advanced), German (intermediate), Chinese (native)
**Programming and Simulation**: Python, C/C++, MATLAB/Simulink
**Hardware**: Arduino, ESP32, Raspberry Pi, NVIDIA Jetson Nano/Xavier; soldering, PCB design and debugging
**Frameworks**: ROS, TensorFlow, PyTorch, TensorRT, YOLO, LangChain, LangGraph, Ollama, llama.cpp
**Development Tools**: Git, Docker

## Selected Coursework (Grades 1.0–1.3; 1.0 Best)

**Robotics**: 1.0 | **Applications of Robotics and Autonomous Systems**: 1.0
**Sensor Systems**: 1.3 | **Machine Learning 1**: 1.3 | **Motion Planning**: 1.3
**Numerical Analysis I for the Engineering Sciences**: 1.0